

# Time Series Forecasting — Engineering Report

Project 009 · ARIMA Forecasting for Datacenter Power & Silicon Aging

## Project Overview

This project implements **ARIMA time series forecasting** across two domains: datacenter power demand prediction (17,520 hourly readings) and silicon transistor aging prediction (1,095 daily Vth shift measurements). The pipeline includes seasonal decomposition, stationarity testing (ADF), ACF/PACF analysis for parameter selection, and multi-step forecasting.

## Datasets

| Dataset                 | Rows   | Frequency | Target       | Domain                    |
|-------------------------|--------|-----------|--------------|---------------------------|
| Datacenter Power Demand | 17,520 | Hourly    | power_kw     | Cloud Infrastructure      |
| Silicon Aging Vth Shift | 1,095  | Daily     | vth_shift_mv | Semiconductor Reliability |

## Methodology

- **Decomposition:** Trend + Seasonal + Residual extraction
- **Stationarity:** ADF test, differencing (d=1)
- **Parameter Selection:** ACF/PACF analysis → ARIMA(5,1,0)
- **Model:** statsmodels ARIMA with maximum likelihood estimation
- **Evaluation:** AIC (Akaike Information Criterion)
- **Serialization:** joblib export (arima\_datacenter.pkl, arima\_silicon\_aging.pkl)

## Results

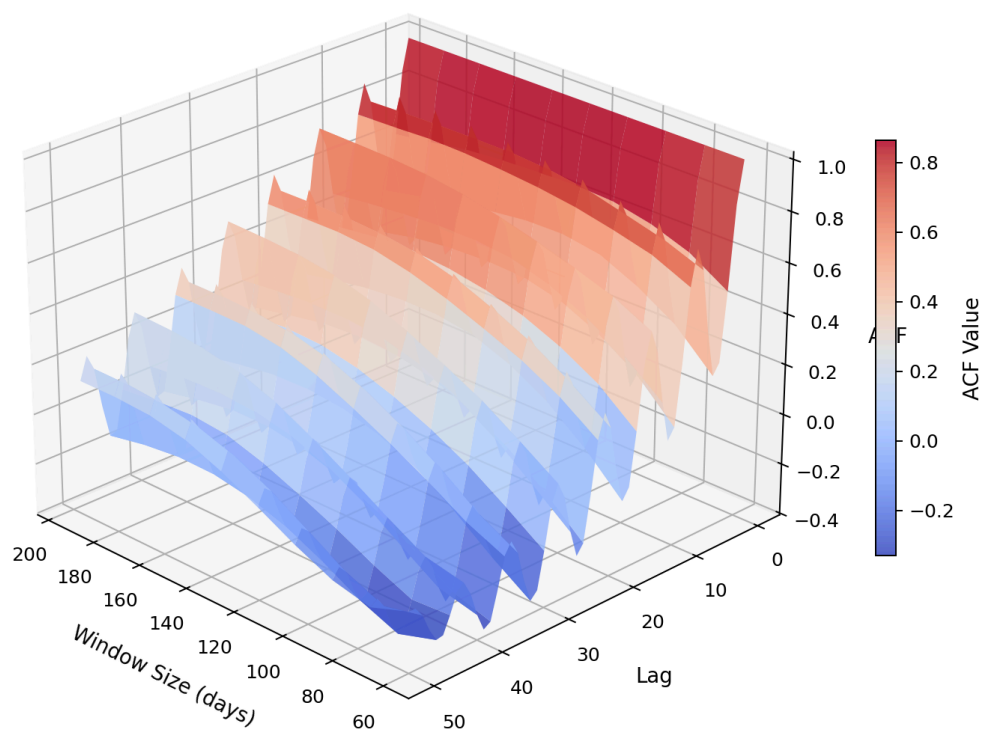
**Datacenter AIC:** 184,514

**Silicon Aging AIC:** 1,772

Lower AIC indicates better model fit relative to complexity. The silicon aging model achieves much lower AIC due to smoother, more predictable degradation patterns.

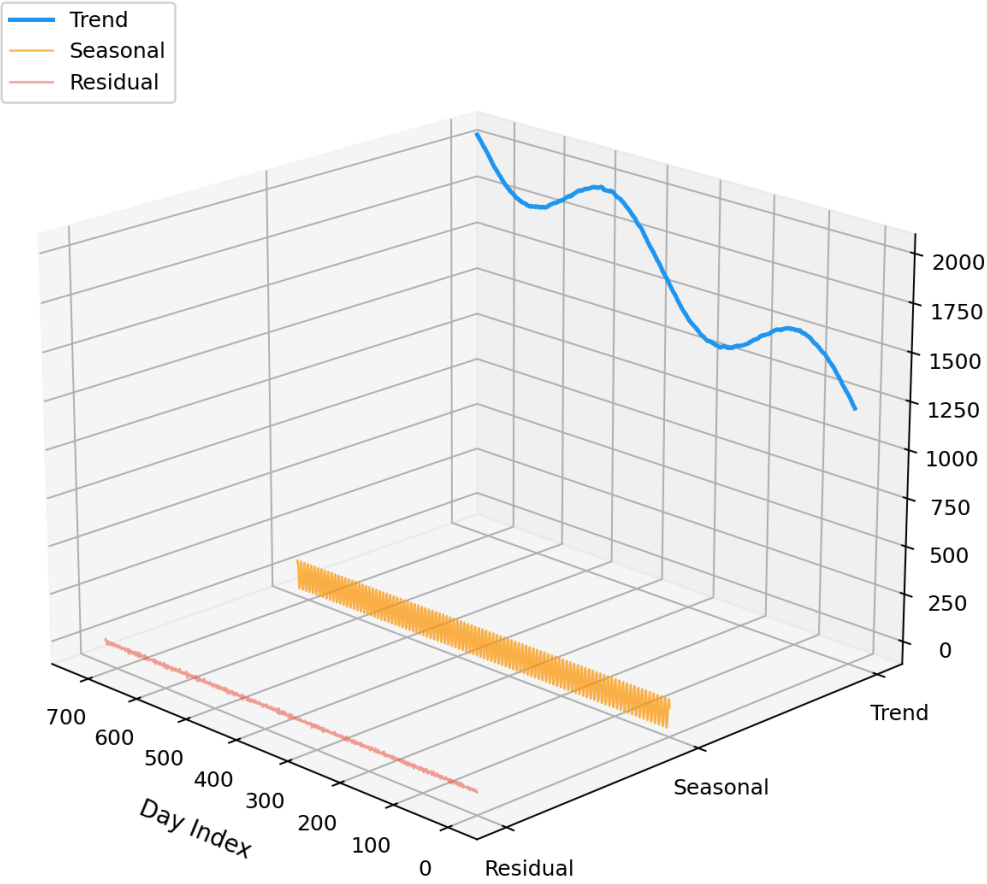
## Project 1 — Datacenter Power Demand Visualizations

### Autocorrelation Function Landscape Varying Window Sizes



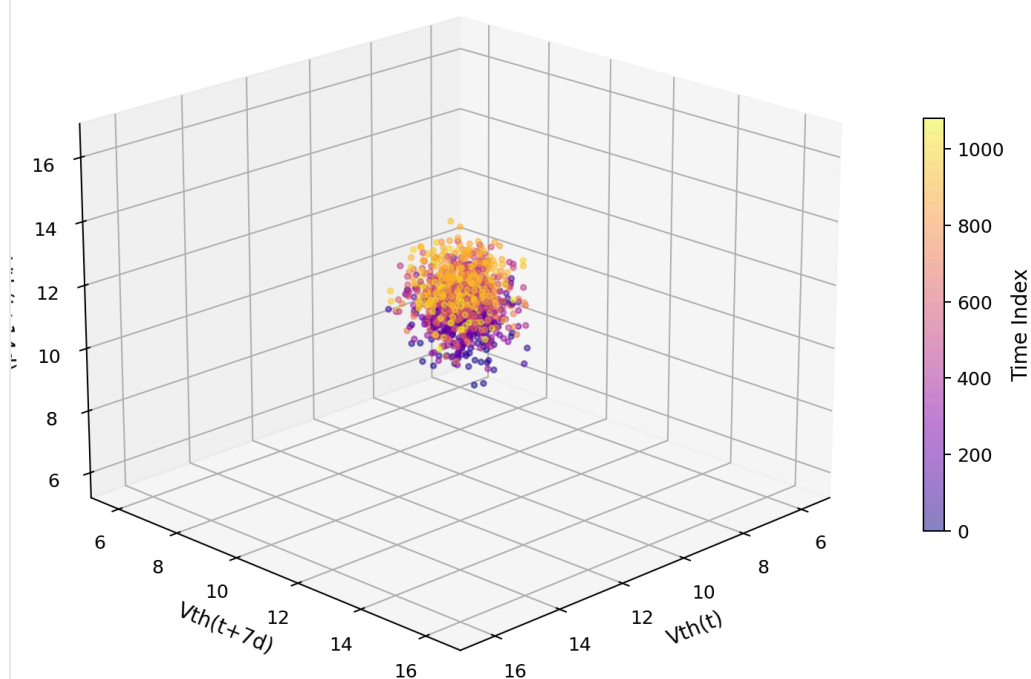
Proj1 Datacenter 3D Acf Landscape

# 3D Time Series Decomposition Datacenter Power Demand

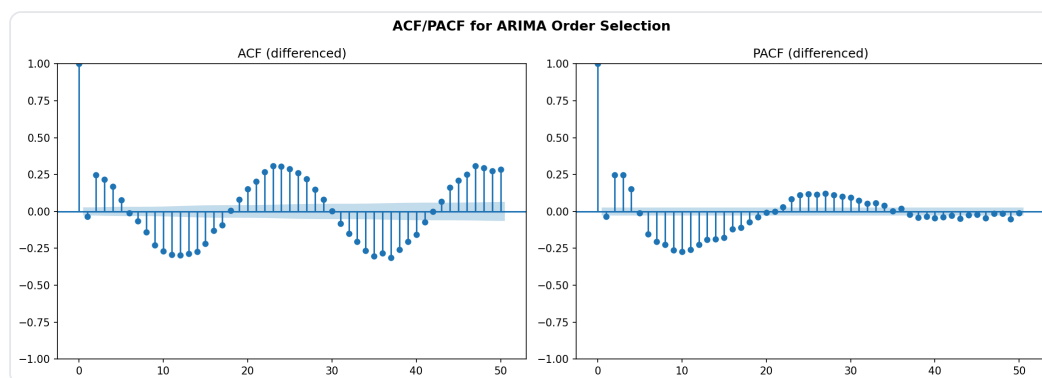


Proj1 Datacenter 3D Decomposition

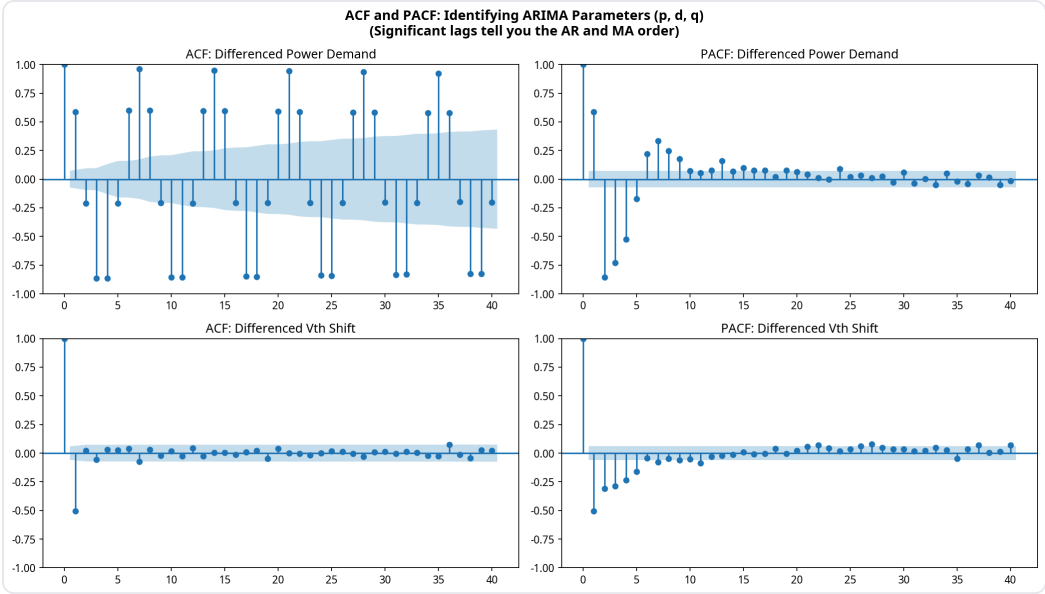
### Silicon Aging Phase Space Embedding Takens' Theorem Reconstruction



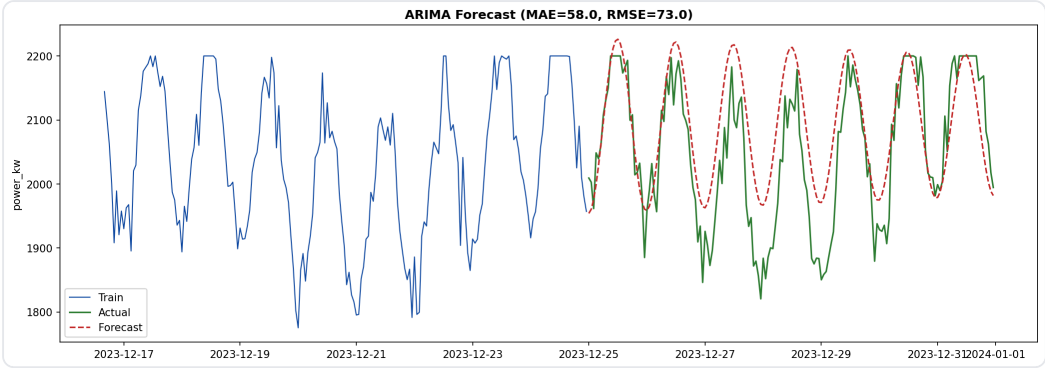
Proj1 Datacenter 3D Phase Space



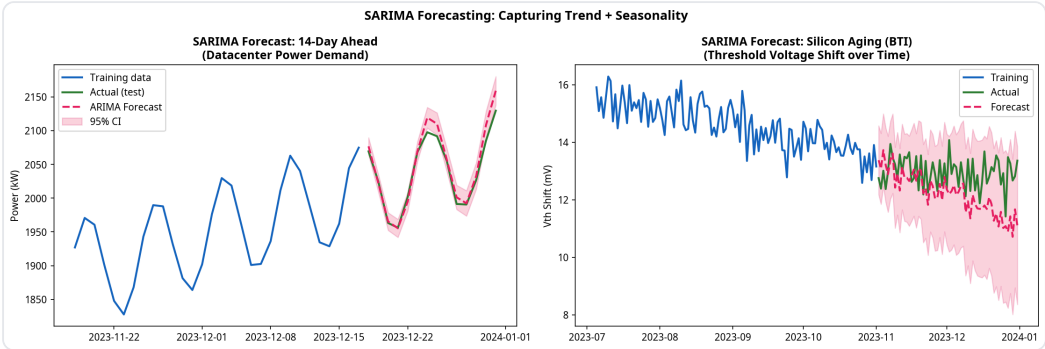
Proj1 Datacenter Acf Pacf



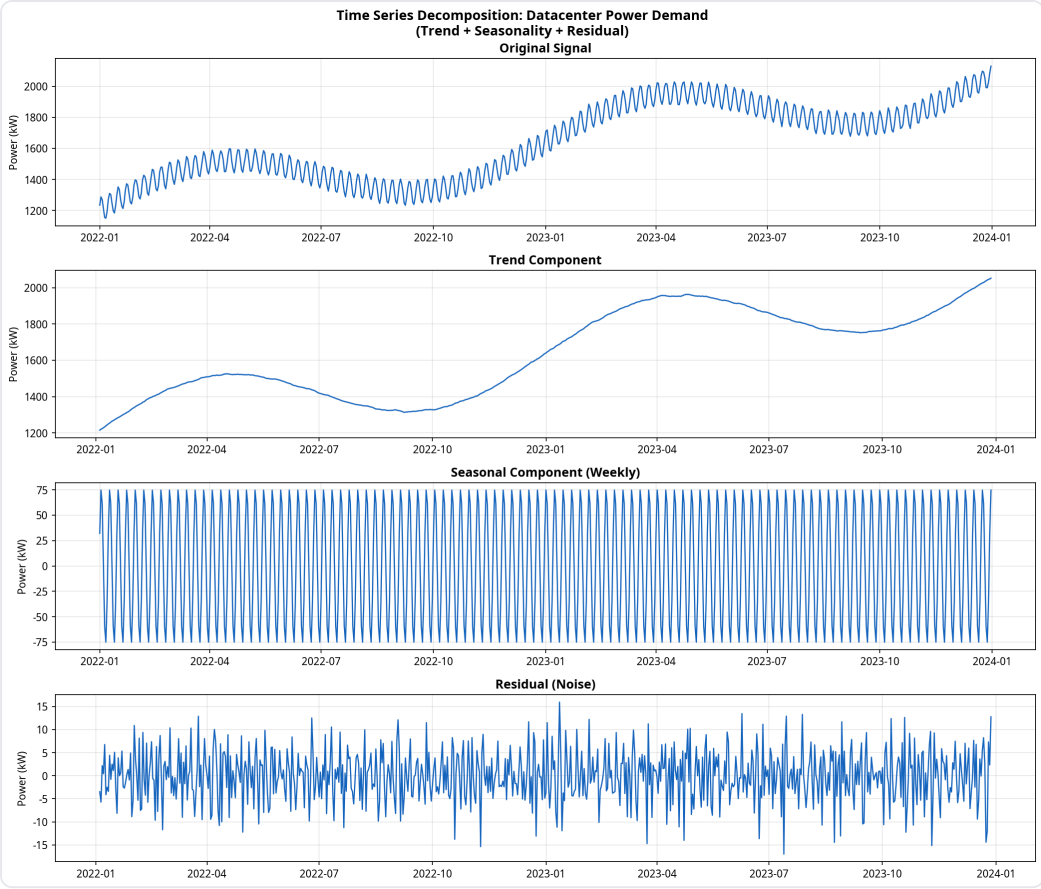
Proj1 Datacenter Acf Pacf Analysis



Proj1 Datacenter Arima Forecast



Proj1 Datacenter Arima Forecast Comparison



Proj1 Datacenter Decomposition

# DATACENTER POWER DEMAND ARIMA FORECASTING PIPELINE



## Time Series Data Ingestion

17,520 hourly power\_kw readings



## Exploratory Data Analysis

Trend Visualization & Seasonal  
Decomposition (trend + seasonal  
+ residual)



## Stationarity Testing

ADF Test & Differencing  
for Stationarity



## ACF / PACF Analysis

Lag Analysis → ARIMA  
Order Selection (p,d,q)



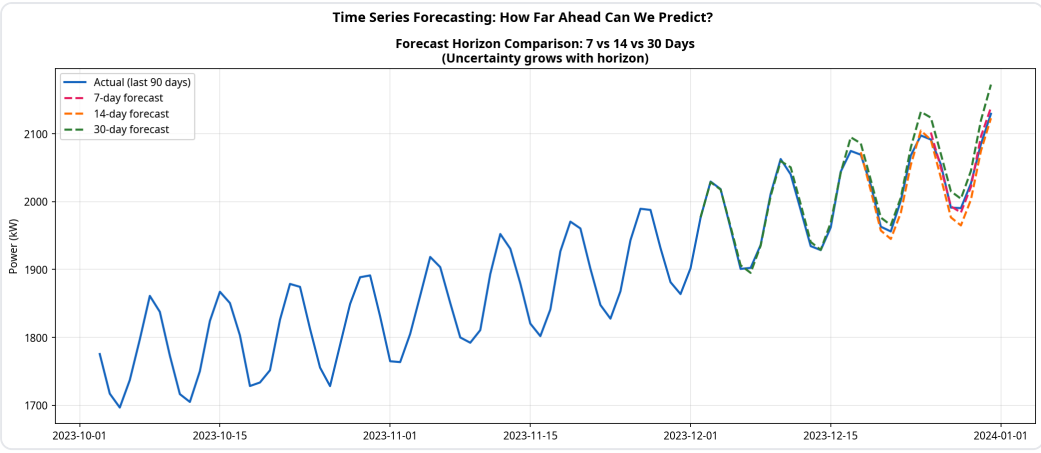
## Forecasting & Evaluation

Multi-step Forecasting,  
Model Selection (AIC=184514)

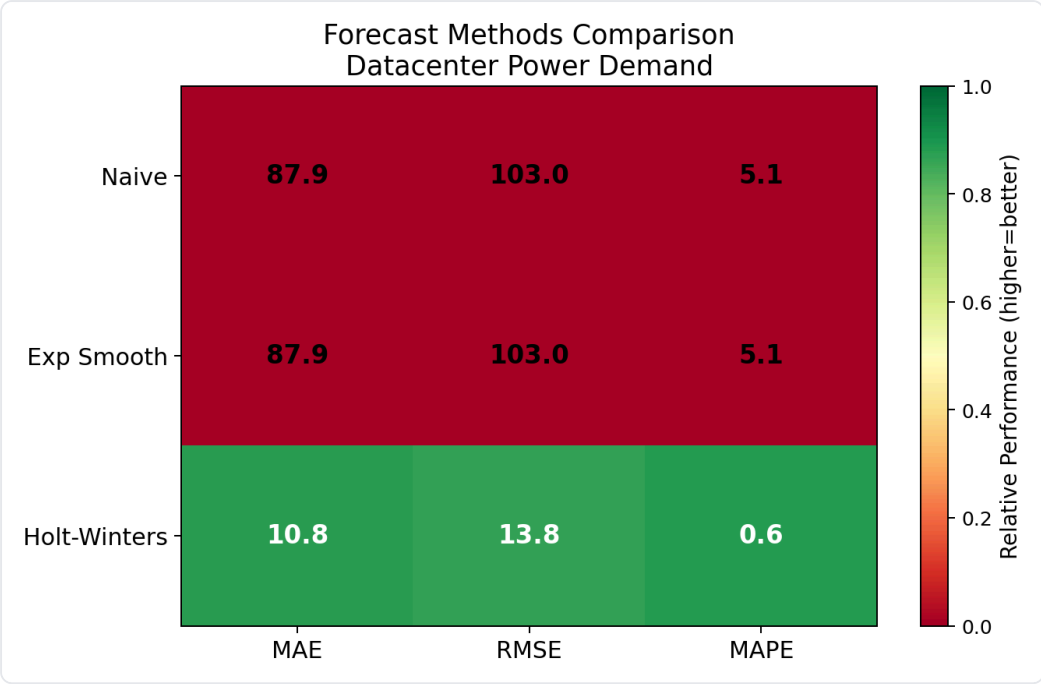


## Model Serialization

Model Saved (joblib for deployment)

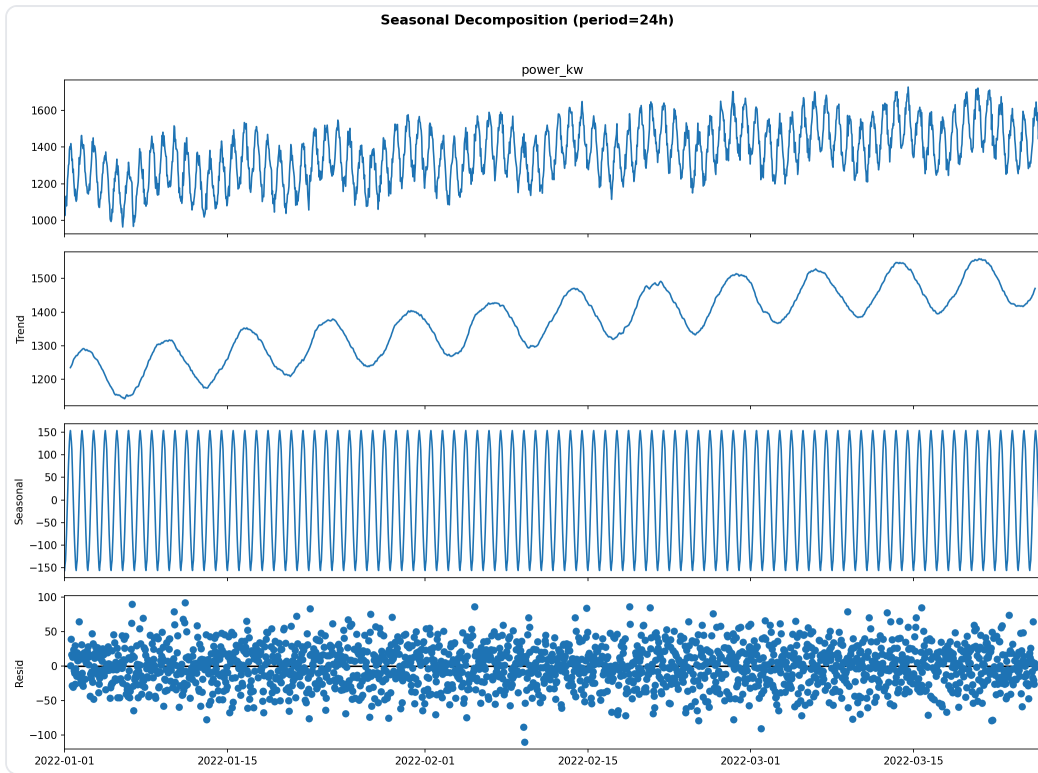


Proj1 Datacenter Forecast Horizons

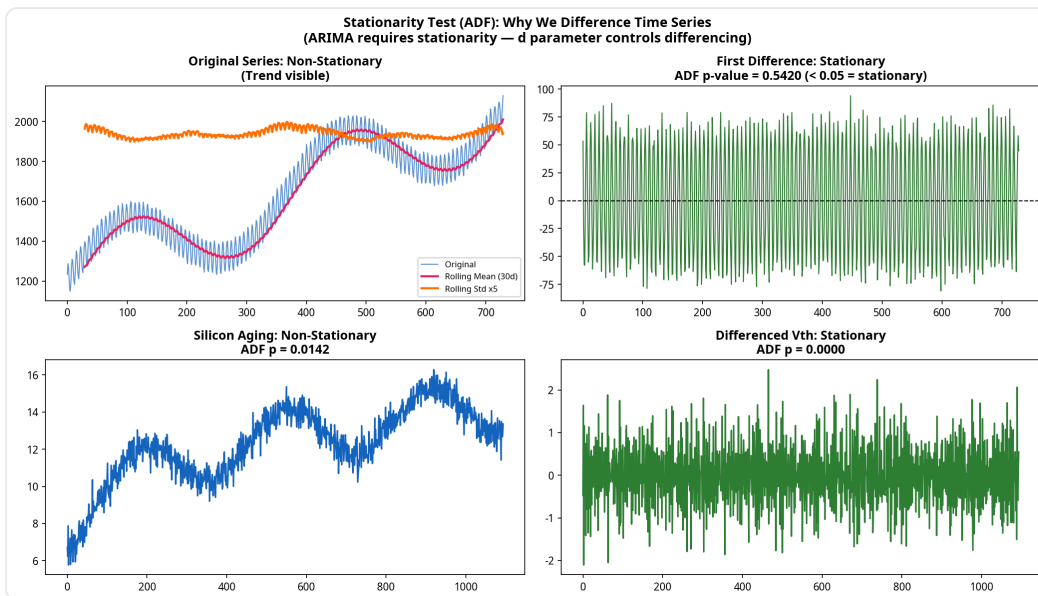


Proj1 Datacenter Model Heatmap

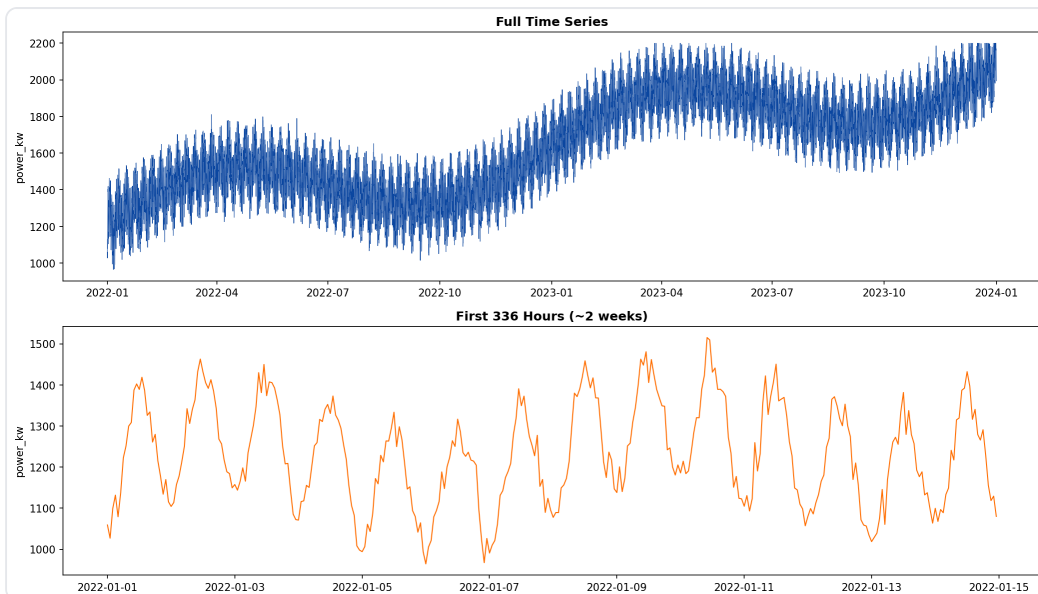




Proj1 Datacenter Seasonal Decomposition

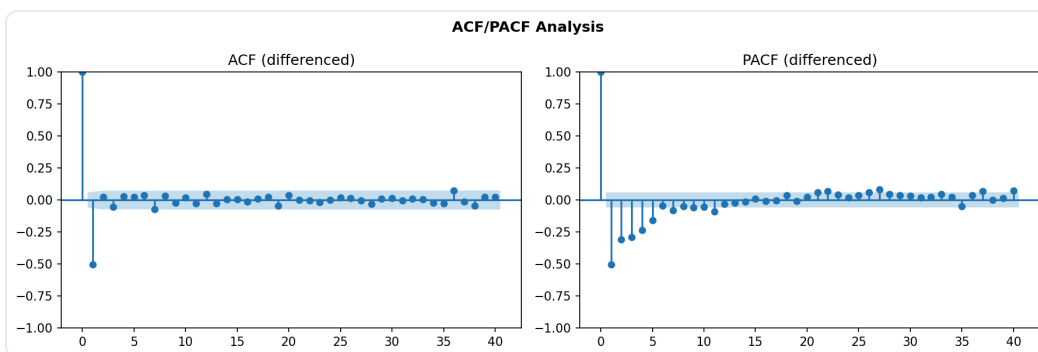


Proj1 Datacenter Stationarity

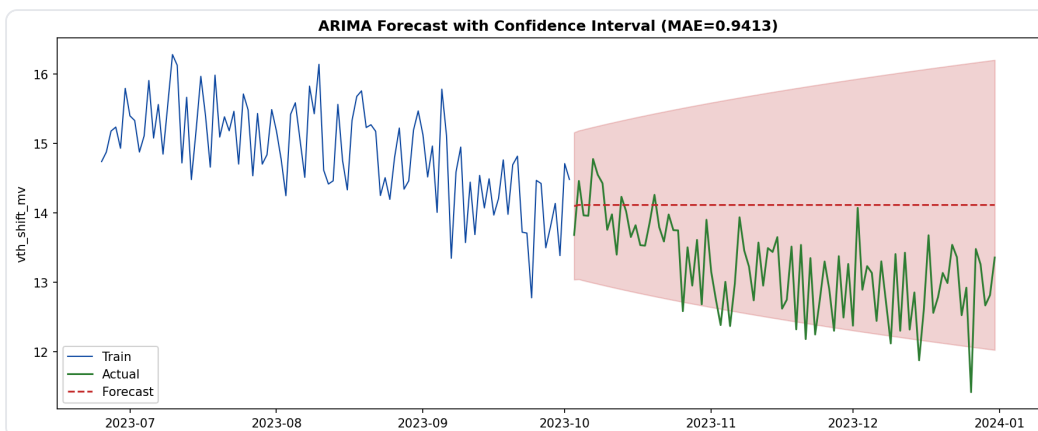


Proj1 Datacenter Ts Overview

## Project 2 — Silicon Aging Vth Shift Visualizations

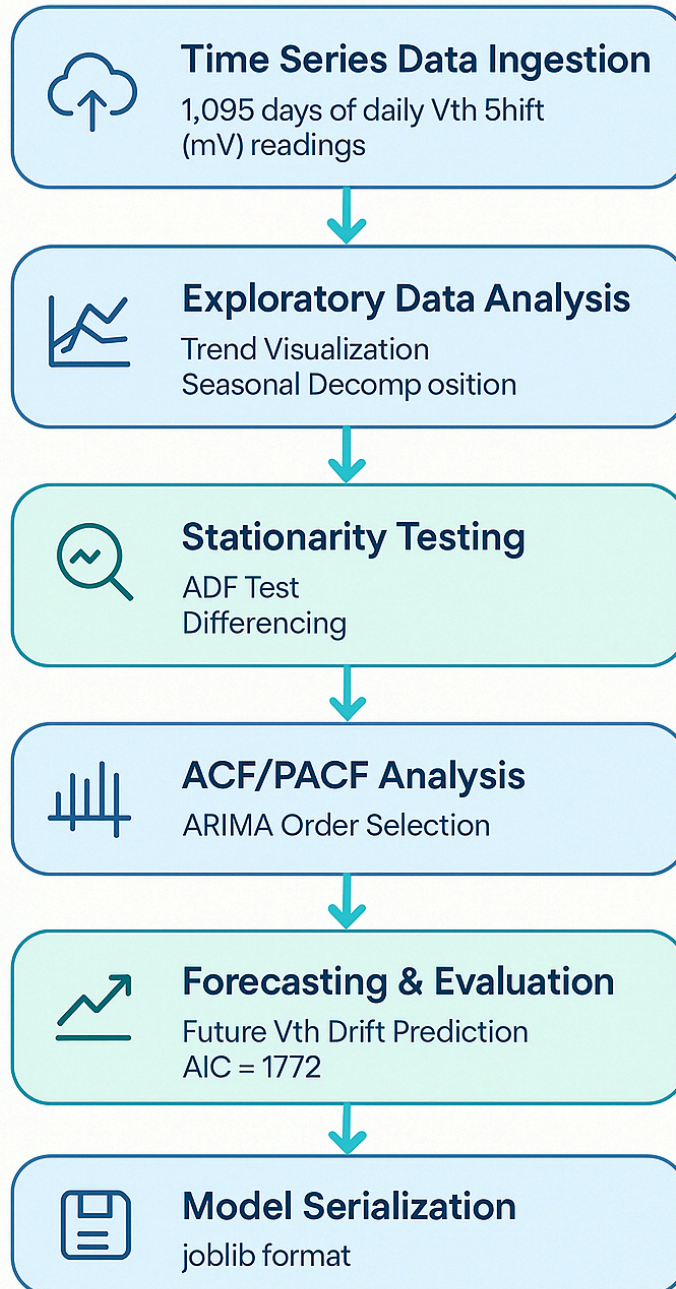


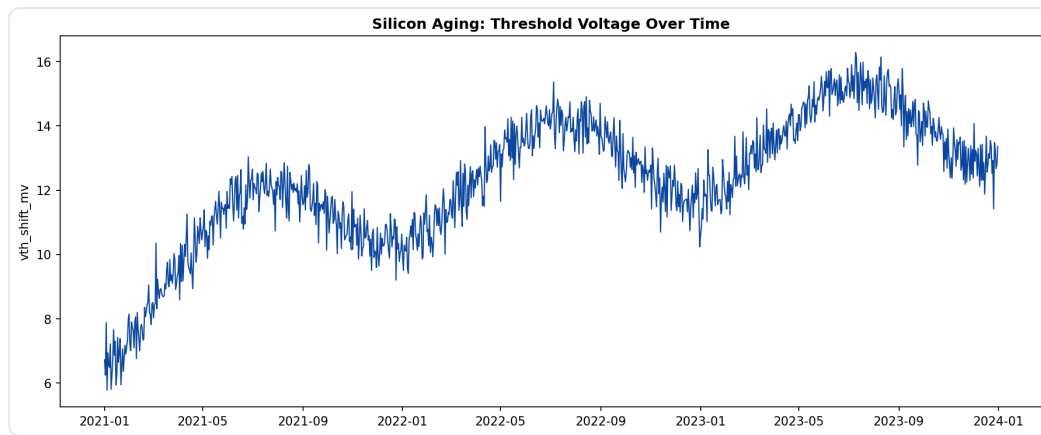
Proj2 Silicon Aging Acf Pacf



Proj2 Silicon Aging Arima Forecast

# Silicon Aging: Vth Shift ARIMA Forecasting Pipeline





Proj2 Silicon Aging Ts Overview

## Architecture

- **Training:** `src/train.py` — DATASETS dict pattern, trains both ARIMA models
- **Inference:** `src/predict.py` — multi-step forecast from any trained model
- **API:** `src/api.py` — FastAPI endpoint, POST steps → forecast values
- **Testing:** `tests/test_model.py` — 4 tests (existence + forecast × 2)